

CreatorLens: 10-Day Hackathon Execution Plan

Revised all-team timeline, task map, scope controls, judging alignment, and submission checklist

Internal build window July 21-31, 2026	Submission deadline Friday, July 31 at 11:59 PM ET
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Working title: CreatorLens - an AI creative director for short-form video.

This plan assumes a three-person team, a public GitHub repository, IBM Bob as the primary development tool, and a working prototype rather than a production-ready editing suite.

1. Win Condition and Scope Lock

The only acceptable MVP: a user uploads a 30-second to 3-minute video, enters a creative brief, receives timestamped AI feedback, and can apply at least one automatic improvement to generate a revised preview.

Must ship

- Web app with upload, creative-brief form, video playback, analysis status, and results view.
- Backend media ingestion and metadata extraction.
- Speech transcription or a reliable transcription fallback.
- Objective audio/visual measurements such as silence, pacing, volume, scene boundaries, brightness, or blur.
- IBM Granite or watsonx-powered reasoning that converts evidence into structured, timestamped feedback.
- One automatic edit: caption burn-in, silence trimming, or audio normalization.
- Public GitHub repository with the required README sections.
- Public demo or presentation video no longer than 3 minutes.
- Completion certificate from at least one IBM Bob-related SkillsBuild course or webinar for every team member.

Cut immediately unless the MVP is already stable

- Full drag-and-drop nonlinear editing timeline.
- Generative B-roll, music generation, or image generation.
- Multi-user collaboration and authentication.
- Long-form video support.
- Mobile app, browser extension, or plugin ecosystem.
- Multiple edit styles beyond one optional stretch preset.
- Perfect frame-by-frame semantic vision analysis.

2. Team Workstreams

Owner	Primary responsibility	Non-negotiable deliverables
ML Backend Lead	AI reasoning pipeline, schemas, feature aggregation, model integration, API contract, backend quality.	Structured analysis JSON; Granite/watsonx response; timestamped feedback; revised-script endpoint; backend tests.
Frontend Lead	User flow, upload form, player, markers, results dashboard, responsive styling.	Working upload flow; player; clickable timestamps; scorecards; recommendations panel; error/loading states.
Media/Integration Lead	FFmpeg/OpenCV processing, automatic edit, storage, rendering, deployment/integration support.	Media extraction; scene/audio metrics; one renderable automatic edit; preview file; deployment support.
Everyone	Bob usage, certificates, documentation, testing, demo preparation, daily integration.	Bob journal entries; public commits; README contributions; final testing; 3-minute demo.

3. 10-Day Execution Calendar

Date	Main objective	All-team tasks	Exit criteria
Tue Jul 21	Scope and foundation	Lock MVP and project name. Create public repo, issues, branch rules, docs folder, Bob journal, API schema draft, wireframe, and SkillsBuild course plan.	Repo is live. Roles assigned. Task board exists. Everyone has Bob access and starts the required learning.
Wed Jul 22	End-to-end skeleton	Frontend upload page and mock results. FastAPI service. Upload endpoint. Save video. Extract duration/audio/frames. Return a fake analysis payload.	A browser upload reaches the backend and displays a mocked result without manual file moving.
Thu Jul 23	Real media evidence	Implement transcription path, silence/volume analysis, scene boundaries, sampled-frame metrics, and unified evidence JSON.	One test video produces timestamps, transcript segments, and objective measurements in one JSON file.
Fri Jul 24	AI creative director	Connect Granite/watsonx. Enforce structured output. Generate summary, scores, timestamped feedback, recommendations, and revised script. Add retries/fallbacks.	Real evidence becomes valid structured feedback for at least three test videos.
Sat Jul 25	Frontend integration	Replace mocks with live analysis. Add progress states, scorecards, feedback cards, and clickable timestamps. Handle API errors.	User uploads a video and sees real feedback inside the web app.
Sun Jul 26	Automatic improvement	Implement one edit path: caption burn-in, silence trim, or audio normalization. Add approve/apply interaction and rendered preview endpoint.	At least one approved AI recommendation produces a playable revised video.
Mon Jul 27	Integration freeze	Connect all components, standardize file paths and schemas, remove hard-coded values, add basic tests, and select the official demo video.	Complete happy path works on a fresh clone. No major architecture changes after tonight.
Tue Jul 28	Quality and judging polish	Improve UI, copy, evidence explanations, accessibility, and failure states. Add before/after comparison. Test edge cases and file limits.	Demo flow is reliable and visibly addresses technical execution, innovation, fit, feasibility, and impact.
Wed Jul 29	Documentation and public proof	Finish README, architecture diagram, Bob usage section, setup instructions, screenshots, license, sample input, and SkillsBuild certificates.	Public reviewer can understand, install, and evaluate the project from GitHub.
Thu Jul 30	Submission production	Record and edit <=3-minute demo. Upload publicly. Draft challenge page. Verify repository and video in incognito/private browser.	All submission fields are complete; only final proofing remains.
Fri Jul 31	Buffer and submit	Internal code freeze by noon ET. Final smoke test. Confirm each certificate. Submit by 6:00 PM ET target, leaving the official 11:59 PM ET deadline as emergency buffer.	Submission page is published; GitHub and video are publicly accessible; links verified by another teammate.

4. Daily Operating Rhythm

Run the project like a compressed engineering sprint. Overtime only helps when integration happens every day; isolated feature work until the final weekend will sink the project.

Time block	Required activity
Start of day - 15 minutes	Each person states yesterday's result, today's single highest-priority deliverable, blockers, and what Bob will be asked to do.
Midday integration	Merge the smallest working slice. Test frontend-to-backend contracts before continuing. Do not wait for a feature to feel perfect.
End of day - 20 minutes	Run the happy path, update issues, add a Bob journal entry, push commits, and record whether the daily exit criterion was met.
Nightly rule	No uncommitted critical work. Any breaking schema change must be announced and documented before teammates continue.

Definition of done for every task

- ☐ Code runs from the documented command or through the web app.
- ☐ Input and output shape is documented.
- ☐ At least one normal case and one failure case were tested.
- ☐ No secret keys, local absolute paths, or private assets are committed.
- ☐ Bob usage is recorded when Bob materially planned, wrote, reviewed, tested, or debugged the task.
- ☐ The relevant issue is updated and the branch is merged or ready for review.

5. Submission Requirement Checklist

Working prototype and IBM Bob

- ☐ Prototype or proof of concept works end to end.
- ☐ IBM Bob is used as the primary development tool and its role is documented with concrete examples.
- ☐ Bob journal contains dated entries and links to commits, pull requests, issues, or generated artifacts.

SkillsBuild learning

- ☐ Every team member completes at least one IBM Bob-related course or webinar.
- ☐ Every completion certificate is downloaded and stored for submission proof.
- ☐ Names on certificates match the submitted team member details.

Public GitHub repository and README

- ☐ Problem statement.
- ☐ Solution description.
- ☐ AI approach and architecture.
- ☐ Selected challenge theme: Creative Industries.
- ☐ How IBM Bob was used.
- ☐ Setup and run instructions.
- ☐ Public sample or screenshots.
- ☐ No credentials or private data.

Challenge submission page

- ☐ Project and team member details.
- ☐ Public GitHub repository link.
- ☐ Publicly accessible demo or presentation video, maximum 3 minutes.
- ☐ Submission page is published before Friday, July 31, 2026 at 11:59 PM ET.

6. Judging Strategy

Criterion	What judges should see	Evidence to build into the demo/README
Technical Execution	A reliable end-to-end pipeline with genuine AI reasoning, structured data, and a real applied edit.	Architecture diagram; live upload; timestamped feedback; rendered improved preview; tests and error handling.
Innovation	The system acts as a creative director and editor, not merely a transcript chatbot.	Evidence-grounded feedback across text, audio, and visuals; user approves an AI-suggested change.
Challenge Fit	Clear alignment with creative workflows and faster content production.	State the Creative Industries theme early; show how the tool bridges imagination and finished output.
Feasibility	A practical, modular product that could scale without claiming impossible automation.	Short-form scope; human approval; standard FFmpeg/OpenCV tooling; defined model and API boundaries.
Real-World Impact	Creators, students, educators, and small teams can improve videos without expert editing knowledge.	Concrete target user; before/after result; time saved; accessibility features such as captions.

7. Highest Risks and Countermeasures

Risk	Trigger	Countermeasure
Model/API access failure	Credentials, quota, model latency, or unavailable IBM service.	Test access on Day 1. Keep a local structured mock and a clearly documented fallback model path while preserving IBM AI use in the judged flow.
Transcription delays	Speech model setup takes too long.	Use the simplest reliable transcription path first. Do not block all downstream work; develop against stored transcript fixtures.
Frontend/backend mismatch	Fields or file paths change repeatedly.	Freeze JSON schemas by Jul 23. Use example payloads and contract tests.
Rendering takes too long	Automatic edit pipeline is unstable.	Choose one simple edit. Caption burn-in or silence trim is enough. Precompute the official demo output if necessary while retaining a live analysis demo.
Too many features	Team begins B-roll, full timeline, auth, or multi-format support.	Refer to the cut list. New features require unanimous approval and completion of the happy path.
Submission failure	Private repository/video, missing certificate, last-minute upload.	Target submission by 6 PM ET on Jul 31. Verify links in an incognito browser and from another teammate's device.

8. Final Demo Script Structure

Time	Content
0:00-0:20	Problem: creators lose time turning rough video into polished content, especially without professional editing expertise.
0:20-0:40	Solution: CreatorLens analyzes the creative brief, transcript, audio, visuals, and pacing as an AI creative director.
0:40-1:45	Live product flow: upload, analyze, click feedback markers, inspect evidence and recommendations.
1:45-2:20	Approve one recommendation and show the revised preview or before/after comparison.
2:20-2:45	Architecture and IBM technologies, including how Bob drove planning, implementation, testing, and debugging.
2:45-3:00	Impact, feasibility, and closing statement.